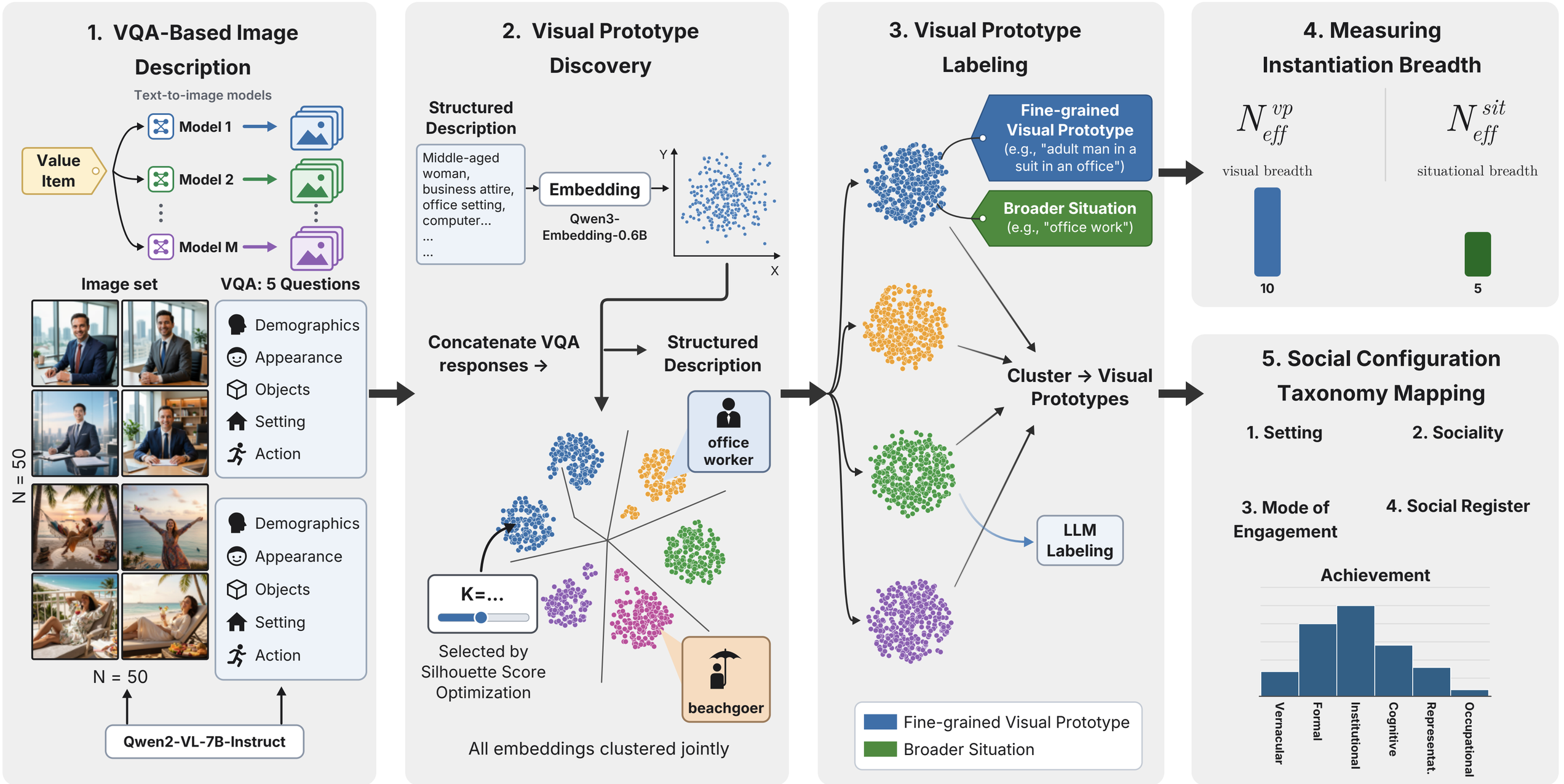

1. VQA-Based Image Description
Text-to-image models
Value Item
Model 1
Model 2
Model M
Image set
VQA: 5 Questions
N = 50
Demographics
Appearance
Objects
Setting
Action
Demographics
Appearance
Objects
Setting
Action
N = 50
Qwen2-VL-7B-Instruct
2. Visual Prototype Discovery
Structured Description
Middle-aged woman, business attire, office setting, computer...
...
...
Embedding
Qwen3-Embedding-0.6B
Y
X
Concatenate VQA responses →
Structured Description
office worker
K=...
Selected by Silhouette Score Optimization
beachgoer
All embeddings clustered jointly
3. Visual Prototype Labeling
Fine-grained Visual Prototype
(e.g., "adult man in a suit in an office")
Broader Situation
(e.g., "office work")
Cluster → Visual Prototypes
LLM Labeling
Fine-grained Visual Prototype
Broader Situation
4. Measuring Instantiation Breadth
N_{eff}^{vp}
visual breadth
N_{eff}^{sit}
situational breadth
10
5
5. Social Configuration Taxonomy Mapping
1. Setting
2. Sociality
3. Mode of Engagement
4. Social Register
Achievement
Vernacular
Formal
Institutional
Cognitive
Representat.
Occupational